B.TECH CSBS MINI PROJECT PRESENTATION

Pick4Me

Hyperlocal Campus Marketplace & Crowdsourced Peer Commute Delivery Platform

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PROJECT GUIDANCE

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Introduction & Campus Marketplace Concept

What is Pick4Me?	Core Purpose & Concept	Target Campus Users
• Hyperlocal community e-commerce and crowdsourced campus peer delivery platform.	• **Monetize Commutes:** Students walk between gates, classes, and hostels daily with empty backpacks.	• **Student Buyers:** Need urgent stationery, lab manuals, medicines, and snacks delivered to their rooms.
• Built specifically for engineering colleges, university campuses, and residential hostels.	• **Digitize Local Vendors:** Canteens, stationary hubs, and Xerox shops gain an online catalog with 0% commission.	• **Student Commuters:** Earn ■20–■50 per delivery during routine walking routes (■200–■500/day).
• Transforms routine student foot traffic into a fast, zero-emission logistics network.	• **Hostel Room Delivery:** Peer commuters deliver directly to rooms, bypassing campus gate bans.	• **Campus Store Merchants:** Expand sales radius across all campus hostels without hiring delivery staff.
• Eliminates high delivery fees and minimum order barriers of commercial apps.	• **Advance Escrow:** Upfront payment locked in escrow eliminates ghosting and buyer disputes.	• **Campus Admin:** Transparent digital monitoring with zero motorized vehicle traffic.

Campus Delivery Challenges & Engineered Solution

Problem Statement: Campus Pain Points

• **High Fees & Minimum Limits:** Commercial delivery apps (Swiggy, Blinkit, Zepto) charge ■30–■60 fees on low-cost essentials (■10 pen, Xerox, lab record), making small orders impractical.

• **Campus Gate Restriction:** Outside commercial delivery agents are prohibited from entering campus hostel premises, forcing students to walk 1 km to the gate late at night.

• **Invisible Local Campus Stores:** Stationery shops, canteens, and medical stores have zero digital presence; students cannot verify product availability or pricing remotely.

• **Unmonetized Routine Commutes:** Students walk multiple kilometers across campus daily without any structured mechanism to earn micro-rewards for peer assistance.

Proposed Solution: Pick4Me Platform

• **Affordable Distance-Tiered Micro-Fees:** Structured campus pricing starting at ■20 base fee with zero minimum order value, perfectly tailored for student budgets.

• **Direct Hostel Room Delivery:** Peer students with valid campus credentials carry packages directly to hostel rooms, eliminating midnight gate trips.

• **Digital Catalog for Campus Merchants:** Enables campus stationery, canteens, and Xerox kiosks to list inventories with real-time stock indicators at 0% commission.

• **Zero-Carbon Peer Logistics Network:** Converts existing foot and cycle traffic into a green, rapid supply network without fuel-burning motorcycles or fleet overhead.

Core Project Objectives & Functional Features

Project Objectives	Key Functional Features
• **Affordable Micro-Logistics:** Eliminate prohibitive delivery surge fees on small essential student purchases.	• **Multi-Store Catalog & Cart (/cart):** Browse 5 campus store categories, live search, real-time stock tags, and unified multi-item cart checkout.
• **Vendor Digitization:** Provide campus shopkeepers with an intuitive, self-managed digital storefront with zero commission.	• **Custom Errand Generator (/create-request):** Request off-catalog items from any unlisted campus store with custom instructions and estimated budget.
• **Student Financial Independence:** Offer flexible micro-earning opportunities during regular campus commutes.	• **Open Delivery Task Board (/deliveries):** Real-time task board for commuting students to claim open deliveries on their routine route.
• **Guaranteed Transaction Trust:** Implement advance UPI escrow and physical OTP verification to eliminate payment fraud.	• **Dynamic Distance-Based Tiers:** Structured zones: Within Campus (■20), Near Gate (■35), and Outer Town Market (■55).
• **Environmental Decarbonization:** Achieve 100% carbon-neutral deliveries by replacing motorized fleets with walking commuters.	• **Live Order Tracking & 4-Digit OTP:** Real-time 2-stage milestone tracking and handover authentication via secret 4-digit OTP.

End-to-End 3-Party Logistics & 2-Stage Escrow Flow

Step 1: Order Placement	Step 2: Commuter Accepts	Step 3: Store Pickup	Step 4: Customer Handover	Step 5: Reward Release
Customer selects items from catalog, inputs delivery location, and pays total amount via UPI QR into Platform Escrow.	Peer student walking in that direction claims delivery task from the open task board (/deliveries).	Commuter reaches shop and collects items. Merchant confirms handover. Product cost is credited to Merchant Wallet.	Commuter reaches customer room. Customer verifies items and provides their private 4-digit Delivery OTP.	Commuter enters OTP. System cryptographically verifies code and releases Delivery Fee + Speed Bonus into Commuter Wallet.

■■ Two-Stage Escrow Protection Guarantee

100% of payment is locked in Escrow upfront. Shopkeepers are guaranteed product payment upon physical handover (Stage 1), and Deliverers are guaranteed delivery earnings upon OTP verification (Stage 2). No cash handling, zero buyer ghosting, and no customer refusal risk.

System Architecture & Component Technology Stack

Layer / Component	Technology Implemented
Backend Framework	Python 3.11+, Flask 3.0 (MVC Architecture)
Cloud Database	PostgreSQL (Neon.tech Serverless Cloud DB)
Local DB Engine	SQLite3 (pick4me.db) with Auto-Migration
Database Driver	psycopg2-binary + sqlite3 Dual Adaptor
Frontend UI/UX	HTML5, Modern CSS3 (Grid/Flexbox), JavaScript
Audio Synthesizer	HTML5 Web Audio API (Dual Oscillator 440/480Hz)
Security & WSGI	PBKDF2:SHA256, Gunicorn WSGI, Render Cloud

Architectural Design Highlights

• **Model-View-Controller (MVC):** Clean separation of business logic in app.py, relational data schemas in database.py, and modular presentation in Jinja2 templates.

• **Dual-Engine Database Adapter:** Dynamically detects DATABASE_URL for 24/7 cloud persistence via Neon PostgreSQL while supporting local development on SQLite3 with zero code changes.

• **Client-Side Web Audio Engine:** Utilizes HTML5 AudioContext to synthesize native dual-frequency ringtones without heavy audio asset files or external dependencies.

• **Production WSGI Orchestration:** Configured with Gunicorn multi-worker WSGI server for reliable production deployment on Render Cloud paired with GitHub CI/CD webhooks.

Database Relational Schema & 6 Core Modules

1. users (Accounts & Auth)	2. shops (Campus Stores)	3. products (Inventory Items)
• id (PK), name, email (UQ), phone • password_hash (PBKDF2) • role ('customer'|'shopper'|'admin') • wallet_balance, is_verified (KYC)	• id (PK), owner_id (FK -> users) • shop_name, category • address, campus landmark, phone • is_active (Store listing flag)	• id (PK), shop_id (FK -> shops) • name, category, price • description, image path • in_stock (Availability status)
4. cart_items (Shopping Cart)	**5. orders (Logistics Ledger)**	**6. wallet_transactions (Audit)**
• id (PK), user_id (FK -> users) • product_id (FK -> products) • quantity (Number of items) • created_at (Session timestamp)	• id (PK), customer_id, commuter_id, shop_id • product_amount, delivery_fee, total • pickup_otp, delivery_otp • status, feedback_rating, comment	• id (PK), user_id (FK -> users) • order_id (FK -> orders), amount • type ('credit_earnings'|'sale'|'debit') • description (#TXN-xxx reference)

Relational Linkages: users (1) ■■< shops (M) ■■< products (M) ■■< cart_items (M) | users (1) ■■< orders (M) >■■ shops (1) | users (1) ■■< wallet_transactions (M)

Security Architecture, Escrow Protocols & Verification

Advance Escrow Payment Protocol

- Customer pays 100% upfront (Product Cost + Delivery Fee) via dynamic UPI QR gateway.
- Funds are locked in Platform Escrow; neither deliverer nor merchant handles customer cash.
- Eliminates customer refusal, late payments, and order cancellations after purchase.
- Automatic refund guarantee in case of canceled or unaccepted orders.

Physical 4-Digit Handover OTP

- Random 4-digit numeric OTP generated at order creation, visible exclusively to the buyer.
- Customer shares code ONLY after physically receiving and inspecting products at hostel door.
- Deliverer cannot complete order or claim payout without verifying this secret token in the app.
- Completely eliminates false delivery claims and disputes.

Cryptographic User Authentication

- Password Protection: PBKDF2:SHA256 salt-hashed encryption via Werkzeug Security.
- Session Security: Cryptographically signed HTTP-only cookies prevent session hijacking.
- Role-Based Access Control: Strict decoratored guards (customer, shopper, admin).
- Campus Member Verification: Mandatory ID proof check and admin KYC approval toggles.

Immutable Financial Ledger

- Every wallet balance movement is backed by an atomic record in wallet_transactions.
- Unique transaction references (#TXN-xxx) generated for each credit and debit.
- Relational foreign key constraints ensure complete financial auditability.
- Direct withdrawal integration to registered student/merchant UPI VPAs (@upi).

Platform Innovations, Campus Benefits & Live Application

Key Technical Innovations

• ■ **P2P In-App Live Audio Calling:** Built-in Web Audio API dual-oscillator synthesizer (440Hz+480Hz) simulating Indian phone ringing with live timer (00:15), mute mic, speaker toggle, plus direct phone (tel:) and WhatsApp fallbacks.

• ■ **Verified Order Feedback & Ratings:** 1-to-5 star rating widget and review submission unlocked on completed orders, displaying verified badges on driver profiles.

• ■ **Dynamic SLA Speed Bonus:** +■10 auto-reward for beating SLA delivery targets, plus Urgent Rush Delivery (+■15) prioritizing emergency night orders.

• ■ **5 Campus Stores Active:** Stationery, Canteen, Tech, Pharmacy, and Night Mart with 29 active products ready for real demonstrations.

Campus Benefits & Live Access

• **100% Zero-Carbon Mobility:** Utilizes natural student walking and bicycle traffic; eliminates fuel consumption and campus noise.

• **Dignified Student Earnings:** Commuter students earn ■200–■500 daily during regular class walks with zero fixed hours or shifts.

• **Local Merchant Empowerment:** Small campus vendors expand sales radius to all hostels with 0% commission fees.

• ■ **Live Deployed Application:**
Production URL: https://pick4me-l868.onrender.com
Cloud Host: Render Web Service (Gunicorn WSGI)
Database: PostgreSQL 16 on Neon Cloud (24/7 Persistence)
Admin Portal: abhishekkanadje7@gmail.com

KOLHAPUR INSTITUTE OF TECHNOLOGY'S COLLEGE OF ENGINEERING KOLHAPUR (EMPOWERED AUTONOMOUS)

Conclusion & Project Summary

Project Summary & Future Scope

- **Comprehensive Solution:** Pick4Me successfully solves high delivery costs, minimum order restrictions, and campus gate barriers.
- **Proven Model:** Combines local merchant e-commerce storefronts with an advance escrow-protected crowdsourced peer delivery network.
- **Production-Ready:** Fully deployed and functional with live PostgreSQL cloud persistence, in-app calling, and automated OTP settlements.
- **Future Scope:** AI route optimization, campus cycle-sharing integration, and multi-campus expansion across 500+ Indian universities.

Thank You!

Questions & Answers

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